

THE ISLAMIA UNIVERSITY OF BAHAWALPUR

DEPARTMENT OF INFORMATION TECHNOLOGY

FINAL YEAR PROJECT PROPOSAL

OneCart

An AI-Driven Fashion Platform with Multimodal Search and Recommendations

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Abstract

OneCart is a MERN-based fashion e-commerce website using MongoDB, Express.js, React.js and Node.js. Customers can browse products, manage carts, place orders and pay through Stripe, Razorpay or Cash on Delivery. It includes secure login, an admin panel and cloud image storage. Its three AI features are a shopping chatbot, personalised recommendations and visual search, helping users get answers and discover products by interest or image.

1. Introduction

OneCart offers customers a convenient place to explore fashion, purchase products and track orders. Administrators manage products and orders through a separate panel. React.js and Tailwind CSS provide a responsive interface, while AI tools improve support and product discovery.

2. Problem Statement

Online stores can make product discovery difficult, provide limited support and rely on manual order handling. Customers may struggle to find a style or get answers about sizes, prices and delivery. Businesses need a central system for products and orders. OneCart addresses these needs with secure shopping, administration tools and AI-assisted discovery.

3. Project Objectives

Main Objectives

- Build a responsive MERN fashion store.
- Provide secure login, cart, checkout and order tracking.
- Integrate Stripe, Razorpay and Cash on Delivery.
- Create an admin panel for products and orders.
- Add chatbot, recommendations and visual search.
- Deploy and test usability, security and AI results.

Specific Objectives

- Use Cloudinary for images and MongoDB Atlas for data.
- Filter products by category, price and keywords.
- Protect user and admin routes with JWT.
- Secure AI endpoints with rate limits and fallbacks.
- Evaluate chatbot accuracy and search relevance.

4. Scope of the Project

Customer Features

- Browse collections, best sellers and product details.
- Search and filter by keyword, category, size and price.
- Manage cart items and proceed to checkout.
- Register, sign in and track personal orders.
- Ask the chatbot about products, delivery and payment.
- View personalised suggestions or search by image.

Admin Features

- Use secure admin login.
- Add, edit and remove products and images.

- View orders, update status and verify payments.

Backend Features

- REST APIs manage users, products, carts, orders and payments.
- JWT secures access; MongoDB and Cloudinary store data and images.
- Protected AI routes connect all three AI features.

5. System Architecture

Frontend (React.js)

React.js builds the interface; Tailwind CSS styles it, React Router manages pages, and Context API shares state. Payment SDKs support checkout.

Backend (Node.js + Express.js)

Express provides REST APIs. JWT and bcrypt support authentication, while Multer handles uploads and environment variables protect credentials.

Database (MongoDB + Mongoose)

MongoDB stores users, products, carts, orders, payments and chat sessions.

AI Layer (Node.js AI Service)

Gemini or Groq can power chatbot responses grounded in store data. TF-IDF and cosine similarity support recommendations, while purchase patterns identify related items. A pretrained model such as MobileNet or CLIP creates image embeddings for visual search.

Deployment

GitHub manages version control; Vercel and supported cloud services can host the application.

6. Key Modules

- 1. Authentication Module — login, registration, password hashing and admin access.
- 2. Product Management Module — image uploads, product updates, filters and search.
- 3. Cart & Checkout Module — size, quantity, totals and payment options.
- 4. Order Processing Module — payments, order history and status updates.
- 5. Admin Panel — manage products and customer orders.

7. AI-Powered Features

The three AI features improve customer support and product discovery.

1. AI Shopping Assistant

The chatbot answers questions using product and store information from MongoDB. Logged-in customers can check their own orders. When information is missing, it should say so rather than guess.

2. Personalised Product Recommendations

TF-IDF and cosine similarity compare product names, categories and descriptions. Available order data can also identify products frequently purchased together.

3. Visual Product Search

Customers upload a fashion image. A pretrained vision model compares image embeddings and ranks similar catalogue items. Results show images, names and prices; keyword search remains a fallback.

AI Features Summary

AI Feature	Technique	Benefit
AI Chatbot	Language-model API + store data	Quick product and order help
Recommendations	TF-IDF, cosine similarity, order patterns	More relevant product discovery
Visual Search	Image embeddings + cosine similarity	Find similar items from a photo

AI Architecture and Safeguards

Flow: React frontend → protected Express AI route → AI service and store data → response. API keys stay on the server. Passwords and payment details are not sent to AI services, and order data is accessed only through authorised APIs. Caching, rate limits and fallback options control cost and keep shopping available if an AI service fails.

AI Evaluation Plan

- Test about 30 chatbot questions for accuracy.
- Review top-five recommendations and calculate precision@5 when labelled examples exist.
- Check if relevant items appear in the top five visual-search results.
- Measure AI response time, targeting under three seconds.

8. Payment Methods Comparison

Payment Method	Integration	Verification	Notes
Cash on Delivery	Backend API	Not required	Pay on delivery
Stripe	Stripe Checkout	Stripe API	Card payments
Razorpay	Razorpay SDK	Razorpay API	Online payment options

9. Project Timeline (4 Months)

- Month 1
  Planning & Design: gather requirements and design the interface, database, APIs and AI approach.
- Month 2
  Development Phase I: build backend models, authentication, initial frontend and recommendation engine.
- Month 3
  Development Phase II: develop catalogue, cart, checkout, payments, admin panel, chatbot and visual search.
- Month 4
  Testing & Deployment: test integrations and AI results, fix bugs, deploy and prepare documentation.

10. Expected Outcomes

- Responsive store with secure authentication.
- Cart, checkout, order tracking and admin tools.
- Stripe, Razorpay and Cash on Delivery.
- Working chatbot, recommendations and visual search.
- Deployment, evaluation and AI safeguards.

11. Conclusion

OneCart combines the essential features of an online fashion store with three practical AI tools. The chatbot provides store-related help, recommendations support product discovery, and visual search helps customers find similar styles from images. With secure APIs, clear fallback options and a manageable MERN-based architecture, the project demonstrates full-stack development and applied AI within a four-month plan.